

Big Data Analytics of Green Stocks vs. Islamic Stocks: Are These Safe Havens?

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Abstract—Green and Islamic stocks are alternative investment that provide investors with the option to build a diversified portfolio. This study explores the safe haven properties of Green stocks and Islamic stocks to shield wealth by using big data analytics. The data was gathered from selected Asia-Pacific countries, and the observation of this study was 8,488. Daily data was utilized from 6 December 2019 to 29 December 2023. To assess the big data analytics diversification benefits, we employed the method of moments quantile regression (MMQR) as the main methodology. The study shows that (1) Green stocks protect investors during uncertain times. (2) Islamic stocks can optimize the returns during different market conditions. (3) Green stocks provide better safe haven benefits than Islamic stocks during market turbulence. Investors need to understand not only the market sentiments but also the regulatory environment of the country before making an investment decision. This research provides information regarding investment strategies for investors, policymakers, and regulators.

Keywords—Big data analytics, Green stocks, Information, Islamic Stocks, Safe havens, Method of moments quantile regression.

I. INTRODUCTION

Climate change is a pressing and substantial issue that demands immediate action and a robust strategy to bolster economic growth and human progress [1]. The concerns about climate change have substantially grown by raising the investors' interest in investments that are environmentally friendly [2]. Therefore, the emergence of green and ethical finance as an innovative financial approach is crucial. It not only fosters economic growth but also preserves the environment, aligning with the sustainable development goals (SDGs) [3]. As time progresses, investors become more mindful of their investment behavior and seek investment opportunities that are ethical and support sustainability goals [4].

Besides seeking environmentally friendly investment, investors also demand financial products that shield their wealth during economic stress. Searching for new diversification potential in the form of safe haven, hedge, and diversifier assets is necessarily needed [5] in combatting uncertainty. The past uncertain times such as Global Financial Crisis (GFC), the European Debt Crisis, and

COVID-19, have brought great recession to the world economy [6].

Green bonds (GBs) are one of the leading green finance products [7]. Thus, investors have been supporting green projects by financing them through GBs [8]. Prior work [9] investigated GBs against traditional bonds, stocks, and commodities. It was further examined that the safe haven and hedge roles of GBs and Sukuk against stock markets [5]. It was documented that GBs serve as a strong hedge against eleven US stock sectors [11]. Similarly, it was found that GBs have a strong hedge function for many international equity markets in Switzerland and G20 countries [12]. Focusing on the US market, the US GBs and its traditional equity was explored [13], while another study compared the US and China GBs and further analyzed their interconnectedness with conventional equities, bonds, monetary policy, and crude oil [14]. Previous research identified the benefits of diversification of clean energy stocks and GBs for dirty energy stocks [15]. Another study also investigated GBs and green stocks [16], while it was then further explored the interrelationship between clean energy, ESG, and GBs against oil shocks [17]. The portfolio benefits of GBs against conventional and Islamic markets was assessed [18], whereas another researchers studied Islamic stocks, GBs, and other classes of assets [19].

Interestingly, none of the prior works considered green stocks and Islamic stocks concurrently as their main interest in diversification benefits study. Therefore, this study addresses the literature gap as follows: (1) comparing green and Islamic stocks, both variables are concurrently our variable of interest. Investors not only consider green investment but also consider it ethical to shield their wealth and mitigate climate change risk by advocating for environmentally sustainable projects [20]. Faith-based and religious investment has gained attention among finance researchers as it can maximize investment return and align with investors' ethical values [5]. Considering both stocks helps to improve the environment and support SDGs [6]. Nevertheless, previous research mostly focused on GBs, a debt instrument as a sole interest, while considering green stocks is also crucial. (2) Stressing on Asia-Pacific markets is noteworthy because climate change poses the greatest threat to this continent [21]. Asia-Pacific has been crowned as the

area with the most disaster-prone worldwide. In 2017, the displacement of nearly 19 million people was due to extreme weather, and half of it was attributed to this region. The loss for the global economy is expected at 40% between 2015 – 2030 due to climate impact [22]. Hence, addressing the environmental gaps and social economics in Asia-Pacific is essential to achieving the SDGs 2030 agenda [23]. (3) We acknowledge uncertainty factors to provide compelling evidence during unexplored economic stress.

The remaining sections of this article are data and methodology, results and discussions, and conclusions in section II, III, and IV respectively.

II. DATA AND METHODOLOGY

A. Data and sources

We gather data from multiple Asia-Pacific countries, including from China, Japan, South Korea, India, Malaysia, Thailand, Indonesia, and Australia. Data availability is limited and is not well developed for the remainder of other countries in the region.

TABLE I. SUMMARY STATISTICS

Var	Obs	Mean	Std.Dev	Min	Max
<i>Stock indices</i>					
GS	8,488	0.004	0.017	-0.999	0.164
IS	8,488	0.002	0.013	-0.152	0.149
<i>Financial assets</i>					
CS	8,488	0.001	0.013	-0.128	0.155
FX	8,488	0.001	0.005	-0.043	0.045
GB	8,487	0.001	0.284	-10.332	11.999
<i>Economic uncertainty</i>					
VIX	8,488	0.003	0.079	-0.234	0.616
GEPU	8,488	0.003	0.009	-0.032	0.064
EMV	8,488	0.470	2.173	-0.965	40.66

We used the stock indices from Morgan Stanley Capital International (MSCI) for Green (GS), Islamic (IS) and Conventional Stocks (CS). We also used other variables like foreign exchange (FX) and government bond (GB) indexes in our model. All the variables are denominated in US Dollars. The prior mentioned variables are country-level data. Meanwhile, to encompass the uncertainty factor in global-level data, we included the stock market volatility index (VIX), global economic policy uncertainty (GEPU), and equity market volatility infectious disease (EMV) index. Each of the global level data were set into each panel (country-level) data. Daily data was gathered from 6 December 2019 to 29 December 2023. We have 8,488 as the total observation and it is categorized as big data in finance. To treat and minimize the effect of the missing values, we conducted a data interpolation and converted all the values into change values.

Table I of the study documents the summary descriptive of the study. It shows that the standard deviation from the daily

returns of EMV is the highest, while foreign exchange returns indicate the lowest.

B. Method of moments quantile regression (MMQR)

To ascertain the financial attributes of the green and Islamic stocks against other financial instruments and uncertainty factors, we used the method of moments quantile regression (MMQR) [24]. It is effective and provides a comprehensive statistical model to conduct such studies. This model helps to accommodate the heterogeneous panel data and provides reliable results for cross-sectional data with endogenous variables. It provides a relationship between predictors and dependent variables in the various market conditions. The model is expressed as follows:

$$Q_y(\theta/X_{it}) = \alpha_i + \delta_i q(\theta) + X'_{it}\beta + Z'_{it}\gamma q(\theta) \quad (1)$$

Where X_{it} signifies the regressors' vector, $\alpha_i + \delta_i q(\theta)$ denotes the quantile- θ fixed effect for individual country (i).

III. RESULTS AND DISCUSSION

We divided the market condition into three levels: bearish (5% - 10%), normal (50%), and bullish (90% - 95%). The data was being run gradually which divided into six panels: Panel A represents green stocks and financial assets, Panel B shows green stocks and uncertainty, Panel C displays Islamic stocks and financial assets, Panel D indicates Islamic stocks and uncertainty, Panel E is about green stocks, financial assets and uncertainty, and Panel F represents Islamic stocks, financial assets and uncertainty.

TABLE II. ORDINARY LEAST SQUARE RESULTS

DV: Green Stocks - Panel A		DV: Green Stocks - Panel B	
CS	0.899	VIX	-0.023
FX	-0.421	GEPU	-0.082
GB	-0.009	EMV	-0.008
DV: Islamic Stocks - Panel C		DV: Islamic Stocks - Panel D	
CS	0.868	VIX	-0.022
FX	0.009	GEPU	-0.065
GB	0.003	EMV	-0.002
DV: Green Stocks - Panel E		DV: Islamic Stocks - Panel F	
CS	0.898	CS	0.871
FX	-0.421	FX	0.012
GB	-0.009	GB	0.004
VIX	-0.003	VIX	0.002
GEPU	-0.002	GEPU	0.008
EMV	-0.003	EMV	-0.001

The overall results for ordinary least squares (OLS) (see Table II) and quantile regression (see Table III) are consistent. Table III shows that green and Islamic stocks provide diversification benefits over conventional stocks in all market conditions. The findings are in line with earlier studies [25]. Panels A and E results indicate that green stocks provide a strong (weak) safe haven against foreign exchange

(government bonds). On the other hand, Islamic stocks act as diversifiers for foreign exchange and government bonds.

TABLE III. QUANTILE REGRESSION RESULTS

	Quantiles				
	0.05	0.1	0.5	0.9	0.95
DV: Green Stocks - Panel A					
CS	0.874***	0.923***	0.997***	0.921***	0.864***
FX	-0.444***	-0.413***	-0.288***	-0.406***	-0.477***
GB	-0.001	-0.001	-0.001*	-0.003	-0.001
DV: Green Stocks - Panel B					
VIX	-0.032***	-0.027***	-0.010***	-0.019***	-0.021***
GEPU	-0.242	-0.184***	-0.014	0.117***	0.183***
EMV	0.003*	0.001	-0.005	-0.001	-0.001
DV: Islamic Stocks - Panel C					
CS	0.805***	0.842***	0.941***	0.869***	0.848***
FX	0.054	0.053	0.006	-0.005	-0.074
GB	0.002**	0.001**	0.002	0.001	-0.001
DV: Islamic Stocks - Panel D					
VIX	-0.032***	-0.023***	-0.007***	-0.017***	-0.014***
GEPU	-0.229***	-0.164***	-0.001	0.126***	0.187***
EMV	0.003*	0.001	-0.005	-0.005	-0.001
DV: Green Stocks - Panel E					
CS	0.862***	0.913***	0.997***	0.927***	0.876***
FX	-0.449***	-0.412***	-0.289***	-0.405***	-0.478***
GB	-0.001	-0.002	-0.001*	-0.004	-0.001
VIX	-0.003	-0.003	-0.002	0.001	0.003
GEPU	-0.084***	-0.035**	-0.008	0.079***	0.076**
EMV	-0.003	-0.009	-0.001	-0.002	-0.007
DV: Islamic Stocks - Panel F					
CS	0.799***	0.843***	0.941***	0.881***	0.853***
FX	0.056	0.059	0.006	0.022	-0.063
GB	0.002**	0.001*	0.002	0.007	-0.001
VIX	0.003	0.002	-0.001	0.005**	0.005
GEPU	-0.083***	-0.037*	-0.001	0.068***	0.086***
EMV	0.006	0.003	0.004	-0.007	-0.001

^a Note(s): ***, ** stands for 1% and 5% level of significance; The significance in italics implied safe haven.

Green and Islamic stocks are highly attractive as we focus on the uncertainty factors (Panel B and D). It shows that green and Islamic stocks can act as a safe haven during uncertain times and protect the value for investors. However, when combined with other financial assets, the attractiveness of green stocks declines. The results are consistent with earlier studies about other green financial assets [26]. Islamic stocks become strong diversifiers during bullish markets (Panel F).

From these results, we can infer that during uncertain times globally due to the pandemic (COVID-19 and Monkey Pox) and political unrest (Russia-Ukraine war), there is more volatility in the financial markets around the world. During those times, investors seek financial assets that provide them with diversification benefits. The results of our study indicate that green and Islamic stocks have safe haven (diversifier) properties in bearish (bullish) markets [27]. Also, the results show that during the period of infectious disease, these financial assets implied a weak function of diversifier (hedge) during a bearish (bullish) market. Our study's results align with previous works [28].

The results have a practical implication, and they give us the inference that green and Islamic stocks are beneficial, especially during uncertain times. These stocks provide those investors with an option to invest and contribute towards the development of the financial markets of the economy who do not want to invest in the debt market or who wish to invest in ethical projects only. The results also provide information about the dynamics of the Asia Pacific financial markets and give us an idea of how alternative financial instruments are developing in this region.

IV. CONCLUSIONS

Investors should consider investing in Green and Islamic stocks to shield wealth. Green stocks have strong safe-haven and hedge properties, and they can mitigate the risk associated with the varying market conditions, while Islamic stocks can be used to optimize returns. Green stocks' performance during uncertain times is better than other financial stocks, and they are a more promising safe haven than Islamic stocks. The performance of financial assets is highly correlated with the overall economic outlook of the economy. Investors also need to understand the market dynamics and regulatory environment before making an investment decision. Therefore, our research provides information for investment strategy.

As our research focused on green and Islamic stocks as environmentally friendly investments, the direction for future research can be conducted as follows: (1) although we did not consider green bonds the pioneer of green investment, future research highly suggested including green bonds as a debt instrument and comparing them with equity instruments. (2) While green and Islamic stocks are traditional assets, it is also interesting to consider green cryptocurrencies as digital assets. Thus, concurrently comparing traditional and digital green assets would give investors a better perspective and investment strategies. (3) Extending the period of research since the beginning issuance of green assets (bonds, stocks, and cryptocurrencies) and then exploring each role during various economic stress (i.e., Global Financial Crisis (GFC), the European Debt Crisis, COVID-19, Russian – Ukraine war, and Palestine – Israel war) could also benefit investors with dynamic perspective of uncertainties.

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AUTHOR CONTRIBUTIONS

Shinta Amalina Hazrati Havidz – conception, project administration, original draft, methodology, review, validation, and supervision. **Umair Saeed Bhutta** – review, validation, and supervision. **Sheren Susanto** – data collection, methodology, original draft, analysis, and interpretation. **Nicholas Lai Wijaya** – data collection, methodology, original draft, analysis, and interpretation. **Najaf Iqbal** – review, validation, and supervision.

DATA AVAILABILITY STATEMENT

The data used in this study cannot be made available due to commercial reasons.

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